

Scott Basedow

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PROFESSIONAL SUMMARY

Mechanical Engineering and Physics student at Stony Brook University with hands-on experience in precision inspection, CMM programming, RF/microwave component assembly, and 3D CAD design. Background spans technical internship work in a hi-rel manufacturing environment, independent engineering design projects, and a strong foundation in physics and applied mathematics. Seeking mechanical engineering opportunities where precision, design, and analytical skills can contribute from day one.

EDUCATION

Stony Brook University

B.S. Mechanical Engineering & B.S. Physics. Honor Roll.

Stony Brook, NY

Expected May 2029

Suffolk County Community College

A.S. Physics. Dean's List, Honor Roll. GPA: 3.8.

Selden, NY

Connetquot High School / BOCES Electrical

High School Diploma. Two-Year Electrical Trades Program.

Bohemia, NY

SKILLS

Design & Engineering: 3D CAD modeling, assemblies, engineering drawings, mechanical design, dimensioning & tolerancing

Manufacturing & Inspection: CMM operation/programming, dimensional inspection, measurement plans, datum alignment, drawing interpretation, tolerance verification

Software: Fusion 360, Verisurf, MATLAB, Python, Microsoft Excel, Microsoft Office

EXPERIENCE

Hi-Rel Engineer Intern

Summer 2026 – Present

Sector Microwave Industries

- Operate and program CMM equipment in Verisurf for precision inspection of RF/microwave components.
- Create measurement plans and datum alignments for bores, diameters, depths, hole locations, planes, and geometric features.
- Verify dimensions and tolerances against engineering drawings and document pass/fail inspection results.
- Assemble RF/microwave rotor and housing assemblies following engineering drawings and shop procedures.
- Compile qualification data from environmental tests, including thermal cycling, thermal shock, vibration, insertion loss, and return loss, for customer part qualification packages.

Project Support Assistant

Jun 2025 – Sep 2025

R&H Consulting

- Supported engineering and construction operations across multiple active project sites.
- Gained field exposure to HVAC, electrical, and building systems.

PROJECTS

EXODUS Mars Rover Platform

Fusion 360

- Independently designed a Mars rover as a CAD final project, incorporating rocker-bogie suspension, robotic mechanisms, articulated solar panels, and rotating sensors.
- Produced engineering drawings, motion studies, dimensional tolerances, fits, and orthographic views.

Alien Invasion Game

Python

- Developed a 2D arcade-style game using object-oriented programming, incorporating collision detection, scoring, enemy spawning, and progressive difficulty scaling.

ACTIVITIES & LEADERSHIP

Stony Brook Motorsports

Mu Alpha Theta

The National Society of Leadership and Success

Phi Theta Kappa

The Planetary Society

Science After Dark

Volunteer

RESEARCH INTERESTS

- Aerospace & Space Systems, Robotics & Mechatronics, Mechanical Design, Advanced Manufacturing, Precision Engineering